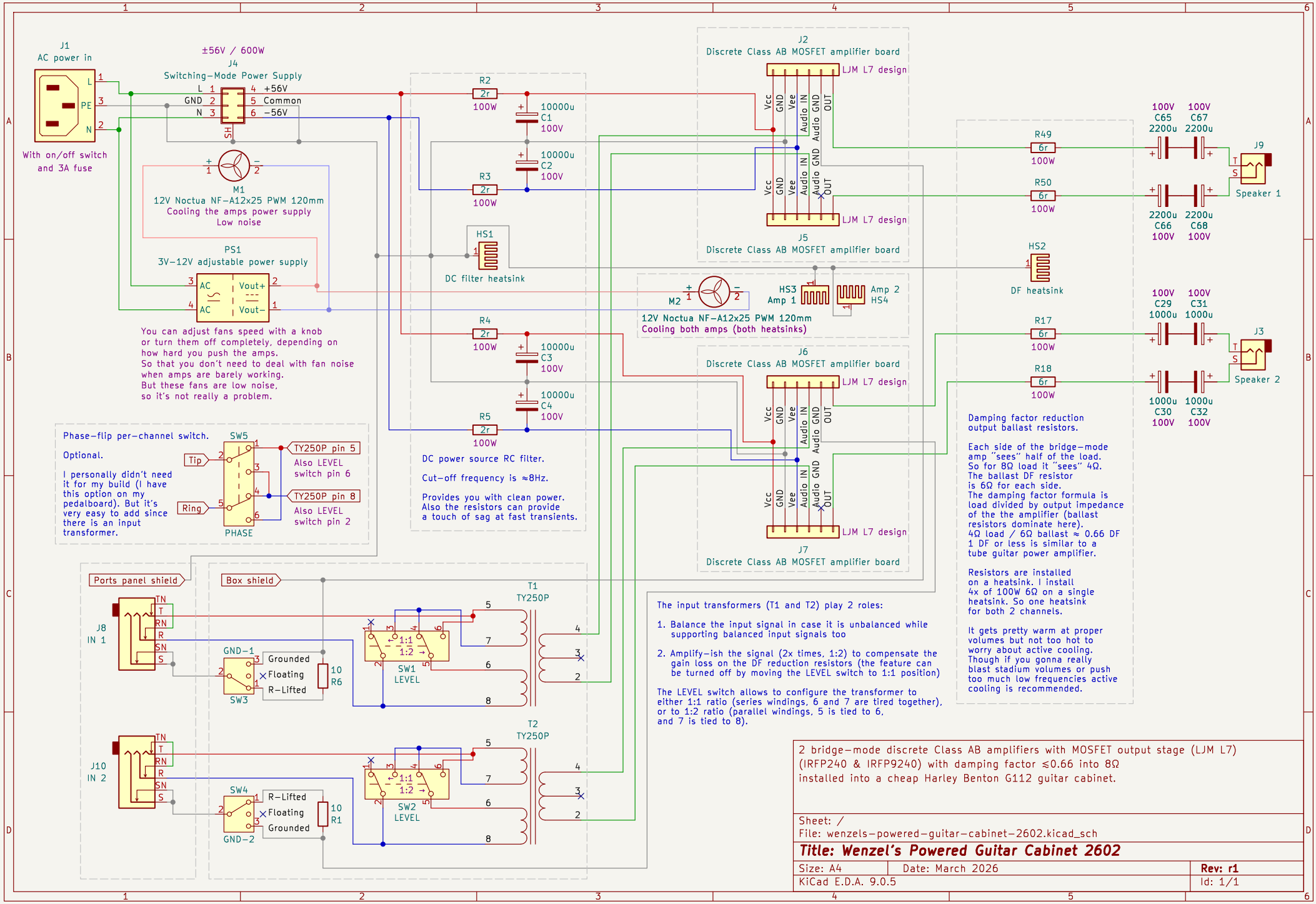
J1
AC power in
With on/off switch
and 3A fuse
±56V / 600W
J4
Switching-Mode Power Supply
L 1
GND 2
N 3
4 +56V
5 Common
6 −56V
SH
M1
12V Noctua NF-A12x25 PWM 120mm
Cooling the amps power supply
Low noise
PS1
3V-12V adjustable power supply
AC
AC
Vout+
Vout−
You can adjust fans speed with a knob
or turn them off completely, depending on
how hard you push the amps.
So that you don't need to deal with fan noise
when amps are barely working.
But these fans are low noise,
so it's not really a problem.
R2
2r
100W
10000u
C1
100V
10000u
C2
100V
R3
2r
100W
HS1
DC filter heatsink
R4
2r
100W
10000u
C3
100V
10000u
C4
100V
R5
2r
100W
DC power source RC filter.
Cut-off frequency is ≈8Hz.
Provides you with clean power.
Also the resistors can provide
a touch of sag at fast transients.
J2
Discrete Class AB MOSFET amplifier board
LJM L7 design
Vcc
GND
Vee
Audio IN
Audio GND
OUT
J5
Discrete Class AB MOSFET amplifier board
LJM L7 design
M2
12V Noctua NF-A12x25 PWM 120mm
Cooling both amps (both heatsinks)
HS3
Amp 1
Amp 2
HS4
J6
Discrete Class AB MOSFET amplifier board
LJM L7 design
J7
Discrete Class AB MOSFET amplifier board
LJM L7 design
R49
6r
100W
R50
6r
100W
HS2
DF heatsink
R17
6r
100W
R18
6r
100W
100V
C65
2200u
100V
C67
2200u
2200u
C66
100V
2200u
C68
100V
J9
Speaker 1
100V
C29
1000u
100V
C31
1000u
1000u
C30
100V
1000u
C32
100V
J3
Speaker 2
Damping factor reduction
output ballast resistors.
Each side of the bridge-mode
amp "sees" half of the load.
So for 8Ω load it "sees" 4Ω.
The ballast DF resistor
is 6Ω for each side.
The damping factor formula is
load divided by output impedance
of the the amplifier (ballast
resistors dominate here).
4Ω load / 6Ω ballast ≈ 0.66 DF
1 DF or less is similar to a
tube guitar power amplifier.
Resistors are installed
on a heatsink. I install
4x of 100W 6Ω on a single
heatsink. So one heatsink
for both 2 channels.
It gets pretty warm at proper
volumes but not too hot to
worry about active cooling.
Though if you gonna really
blast stadium volumes or push
too much low frequencies active
cooling is recommended.
Phase-flip per-channel switch.
Optional.
I personally didn't need
it for my build (I have
this option on my
pedalboard). But it's
very easy to add since
there is an input
transformer.
SW5
Tip
Ring
PHASE
TY250P pin 5
Also LEVEL
switch pin 6
TY250P pin 8
Also LEVEL
switch pin 2
Ports panel shield
Box shield
J8
IN 1
J10
IN 2
TN
T
RN
R
SN
S
GND-1
SW3
Grounded
Floating
R-Lifted
10
R6
SW1
LEVEL
1:1
1:2
T1
TY250P
SW4
GND-2
R-Lifted
Floating
Grounded
10
R1
SW2
LEVEL
T2
TY250P
The input transformers (T1 and T2) play 2 roles:
1. Balance the input signal in case it is unbalanced while
supporting balanced input signals too
2. Amplify-ish the signal (2x times, 1:2) to compensate the
gain loss on the DF reduction resistors (the feature can
be turned off by moving the LEVEL switch to 1:1 position)
The LEVEL switch allows to configure the transformer to
either 1:1 ratio (series windings, 6 and 7 are tired together),
or to 1:2 ratio (parallel windings, 5 is tied to 6,
and 7 is tied to 8).
2 bridge-mode discrete Class AB amplifiers with MOSFET output stage (LJM L7)
(IRFP240 & IRFP9240) with damping factor ≤0.66 into 8Ω
installed into a cheap Harley Benton G112 guitar cabinet.
Sheet: /
File: wenzels-powered-guitar-cabinet-2602.kicad_sch
Title: Wenzel's Powered Guitar Cabinet 2602
Size: A4
Date: March 2026
Rev: r1
KiCad E.D.A. 9.0.5
Id: 1/1